

Quiz 9

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Find the least squares fitting line $y = C + Dt$ for $b = 1, 2, 4$ at times $t = -1, 0, 2$.

Solution: $AX=b$ where $A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, $x = \begin{bmatrix} C \\ D \end{bmatrix}$

We solve

$$A^T A \hat{x} = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{cases} C=2 \\ D=1 \end{cases}$$

Thus the least squares fitting line is

$$y = 2 + t$$

Problem 2. (6 points) Find an orthonormal basis $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ for the column space of

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix} \text{ using the Gram-Schmidt process.}$$

Solution: $\tilde{\mathbf{q}}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \|\tilde{\mathbf{q}}_1\|^2 = 1$

$$\tilde{\mathbf{q}}_2 = \mathbf{a}_2 - \left(\frac{\tilde{\mathbf{q}}_1^T \mathbf{a}_2}{\|\tilde{\mathbf{q}}_1\|^2} \right) \tilde{\mathbf{q}}_1 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{2}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}, \|\tilde{\mathbf{q}}_2\|^2 = 9$$

$$\tilde{\mathbf{q}}_3 = \mathbf{a}_3 - \left(\frac{\tilde{\mathbf{q}}_1^T \mathbf{a}_3}{\|\tilde{\mathbf{q}}_1\|^2} \right) \tilde{\mathbf{q}}_1 - \left(\frac{\tilde{\mathbf{q}}_2^T \mathbf{a}_3}{\|\tilde{\mathbf{q}}_2\|^2} \right) \tilde{\mathbf{q}}_2$$

$$= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \left(\frac{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}}{9} \right) \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \left(\frac{4}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{18}{9} \right) \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \|\tilde{\mathbf{q}}_3\|^2 = 25$$

Thus

$$\mathbf{q}_1 = \frac{\tilde{\mathbf{q}}_1}{\|\tilde{\mathbf{q}}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{q}_2 = \frac{\tilde{\mathbf{q}}_2}{\|\tilde{\mathbf{q}}_2\|} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} / 3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{q}_3 = \frac{\tilde{\mathbf{q}}_3}{\|\tilde{\mathbf{q}}_3\|} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} / 5 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$